

CSE 220: Signals and Systems Sessional

Online – Continuous Fourier Transform

Time: 30 Minutes

Definitions

For a CFT spectrum of size $H \times W$, the center is $(c_i, c_j) = (\lfloor H/2 \rfloor, \lfloor W/2 \rfloor)$. Let $d(i, j) = \sqrt{(i - c_i)^2 + (j - c_j)^2}$ be the Euclidean distance of entry (i, j) from the center.

Task 1: Band-Pass and Band-Stop Filters

Implement the following filters in the `FrequencyFilter` class. Both apply to the real and imaginary spectrum arrays.

- **Band-Pass:** Retain entries where $r_{\text{low}} < d(i, j) \leq r_{\text{high}}$, zeroing the rest.
- **Band-Stop:** Zero entries where $r_{\text{low}} < d(i, j) \leq r_{\text{high}}$, retaining the rest.

Task 2: Complementarity Check

Since the two filters above are exact complements, their reconstructed signals satisfy:

$$I_{\text{bp}}(x, y) + I_{\text{bs}}(x, y) \approx I_{\text{recon}}(x, y)$$

In `ReconstructionValidator`, check this property on the raw inverse-CFT arrays by computing the maximum absolute difference:

$$\delta = \max_{x, y} |I_{\text{bp}}(x, y) + I_{\text{bs}}(x, y) - I_{\text{recon}}(x, y)|$$

Return `True` if $\delta < 10^{-9}$, and `False` otherwise, along with the computed δ .

Task 3: DC Component Manipulation

The exact center of the frequency spectrum (c_i, c_j) corresponds to the average brightness (DC component) of the entire image.

In `FrequencyFilter`, implement `shift_brightness` by adding `shift_amount` strictly to the real component of this center pixel, leaving all other components unchanged.